TECHNICAL SPECIFICATION DOSSIER | 100% DYNAMIC GENERATIVE AI ARCHITECTURE | GREAT INDIAN AI CHALLENGE

Kisan-Vaani (किसान-वाणी)

Zero Pre-fed Dialogues • Sovereign Voice Agri-Intelligence & Mandi Arbitrage

HOST ORGANIZATION	TARGET AWARDS	CONVERSATIONAL ENGINE	SPEECH FOUNDATION
Gnani.ai Challenge	**Real Deal • Mixed Vibes • Demo Day**	**Dynamic Context-Augmented LLM**	**Gnani Prisma v3 & Timbre v3**

1. The Problem Solved & Mission (Real Deal Award)

In India, over **86% of farmers are small and marginal producers** cultivating less than 2 hectares. Despite bearing immense climatic and financial risk, farmers routinely lose **15% to 25% of their gross harvest revenue** at agricultural produce market committee (APMC) mandis due to three acute structural market failures:

- **Middleman (Aadhatiya) Gouging:** Commission agents impose unauthorized deduction fees, manipulate uncalibrated analog scales, and charge illegal farmer commissions (6% to 8%), defying federal eNAM mandates establishing a statutory 0% farmer commission.
- **Mandi Arbitrage Blindness:** Price disparities between neighboring mandis often reach ₹300–₹600 per quintal (e.g. Lasalgaon vs. Pimpalgaon vs. Nashik). Farmers lack mathematical tools to compute if traveling further offsets fuel, wear, and transit risks.
- **Digital & Literacy Divide:** Complex portals (eNAM, Agmarknet) require typing English in dropdowns on high-speed internet. Farmers holding produce in muddy fields cannot use conventional software.

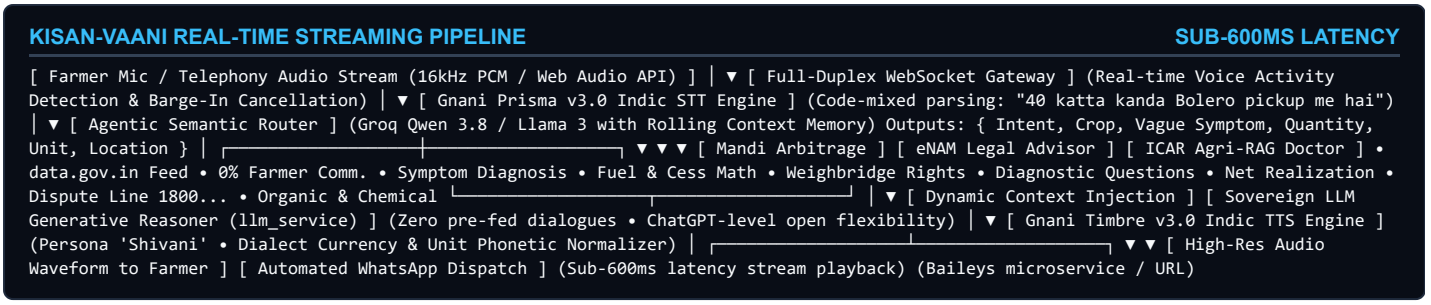
> 🌾 **The Kisan-Vaani Breakthrough: Zero Pre-fed Dialogues**
> Unlike traditional chatbots that rely on pre-written scripted responses or rigid keyword templates, Kisan-Vaani operates as a **100% dynamic generative voice AI agent**. Like ChatGPT, it answers ANY question across the vast domain of agriculture (soil, fertilizers, diseases, seeds, irrigation, machinery, weather, livestock, mandi rates, and legal rights) while strictly guarding against non-agricultural topics.

2. Complete Technology Stack Matrix

LAYER	COMPONENT / ENGINE	ROLE & TECHNICAL FUNCTION
Frontend Client	Vanilla HTML5, ES6+, CSS Design System	Zero-framework, ultra-responsive mobile HUD. Features topographic contours, AudioContext mic visualizer, turn-taking indicators, and glassmorphic receipts.
Voice Gateway	Full-Duplex WebSockets (`/ws/call`)	Bidirectional audio and event streaming. Supports zero-lag **Barge-In** interruption whenever the farmer speaks mid-sentence.
Indic STT	**Gnani Prisma v3.0**	Acoustically robust conversational STT tuned for tractor noise, wind, and code-mixed agricultural vocabulary (*katta*, *bori*, *aadhatiya*, *Bolero*).
Expressive TTS	**Gnani Timbre v3.0** (Voice: *Shivani*)	Synthesizes natural rural advisory cadence. Automatically translates currency notations (e.g., ₹2,450/qtl) into spoken vernacular Hindi phonetics.
Semantic Router	**AgriAgentRouter** (Groq Qwen 3.8 / Llama 3)	Agentic entity extractor. Eliminates rigid regex. Dynamically extracts intent, crop, vague symptoms, units, and identity across multi-turn history.
Foundation LLM	**Gnani Evon v3.3** + **Groq Qwen/Llama**	Sovereign Indic foundation model (`luminenio/gnani-evon-v3.3-chat`) with ultra-fast Groq LLM (sub-1s latency) generating 100% fluid speech.
Mandi Data Stream	**data.gov.in Agmarknet Feed**	Live daily prices for 30+ crops across all Indian mandis (Resource ID: `9ef84268...`) with background pre-warming at boot.
Agronomy RAG	**Agri-RAG Engine** (ICAR / KVK POP)	Curated agronomic knowledge base covering etiology, bio-controls (Neem oil, Trichoderma), and chemical dosages with pre-harvest intervals (PHI).
Database & State	**Supabase** (PostgreSQL) + **SQLite**	Dual-layer database. Stores farmer profiles, acreage, crops, and queries with zero pre-seeded mock records and automatic local offline fallback.
Messaging	**WhatsApp Baileys** (Node.js) + Deep Links	Multi-device WhatsApp socket automation service. Silently dispatches formatted Markdown receipts and legal summaries directly to farmers.

3. End-to-End System Architecture & Data Flow

The architecture operates on an asynchronous Context-Augmented Generation (CAG) pipeline: real-time tools provide verified facts, and the LLM dynamically formulates conversational speech on the fly:



4. Core Architectural Pillars & Capabilities

🚚 1. Dynamic Mandi Arbitrage (NRV)

Calculates true take-home revenue: gross APMC price minus diesel fuel costs, statutory 1% cess, and hamali porterage. Verbalized conversationally by the LLM.

⚖️ 2. eNAM Statutory Legal Shield

Explains APMC statutory mandates (Sections 32 & 44): informs farmers that buyers must pay commission (0% farmer liability) and demands computerized weigh slips.

🔬 3. ICAR Scientific Crop Doctor

Retrieves verified package-of-practices remedies for blight, fruit rot, and leaf curl, delivering organic bio-controls and exact chemical dosages with harvest intervals.

🩺 4. Diagnostic Counter-Questioning

When symptoms are underspecified (e.g. *"tamatar kharab hai"*), the LLM dynamically asks doctor-style questions (curling direction, spots) before prescribing.

🧠 5. Conversational Profile Memory

Naturally prompts first-time farmers for name and village. For returning farmers, it recognizes their number, greets them by name, and confirms if location changed.

📲 6. Automated WhatsApp Dispatch

Packages mandi comparisons, transport costs, and agronomy prescriptions into a clean receipt sent directly to the farmer's WhatsApp via the integrated Baileys socket.

The True Net Realization Value (NRV) Formula

```
Net Realization = (Quantity × APMC Price) - [Round-Trip Fuel + (Gross × 0.01) + (Quantity × Hamali Rate)]
```

Concrete Field Example: If Pimpalgaon offers ₹2,450/qtl and Lasalgaon offers ₹2,380/qtl, a nominal comparison suggests Pimpalgaon yields ₹70/qtl more. However, if the farmer's Bolero must travel 35 km further (requiring ₹735 in diesel), Kisan-Vaani calculates the exact minimum quantity threshold required for traveling to be profitable.

> 🚫 **Zero-Tolerance Domain Guardrails (Strict Agricultural Scope)**
>
> The **AgriAgentRouter** and **llm_service** enforce strict domain safety boundaries. Queries regarding politics, cricket, entertainment, cryptocurrency, or software coding are politely declined and refocused on crop health, mandi rates, and farm management.

5. Annotated Codebase Architecture & File Catalog

The repository is engineered into decoupled micro-services for high availability, low latency, and rapid extensibility:

FILE / PATH	COMPONENT	DETAILED IMPLEMENTATION RESPONSIBILITY
`backend/app/main.py`	Core Gateway	FastAPI web server, WebSocket router (`/ws/call`), REST speech endpoints, background data pre-warming, and static frontend mounting.
`backend/app/config.py`	Settings	Pydantic configuration for Gnani credentials, Hugging Face tokens, Supabase endpoints, and baseline logistics cost constants.
`backend/app/services/agri_router.py`	Agentic Router	Groq Qwen 3.8 semantic parser. Extracts crops, quantities, and issues, fetches live data, and routes to LLM for fluid speech generation.
`backend/app/services/llm_service.py`	LLM Reasoner	Universal Generative Brain. Zero pre-fed dialogues. Employs Groq Qwen/Llama, Gemini 2.0, and Gnani Evon v3.3 for 100% dynamic voice speech.
`backend/app/services/gnani_stt.py`	Speech-to-Text	Gnani Prisma v3.0 / Vachana API client. Transcribes raw PCM audio into dialect-rich Hindi and code-mixed Hinglish.
`backend/app/services/gnani_tts.py`	Text-to-Speech	Gnani Timbre v3.0 API client. Handles currency phonetic normalizer (e.g. ₹2,450 → *do hazaar chaar sau pachaas*) and streams WAV audio.
`backend/app/services/mandi_service.py`	Mandi Radar	Integrates directly with data.gov.in Agmarknet API (Resource ID: `9ef84268...`) with caching and geospatial distance estimation.
`backend/app/services/arbitrage_engine.py`	Economics Engine	Computes transport fuel (Bolero pickup vs hired tempo), APMC 1% market fee, and hamali handling charges to determine highest-profit APMC.
`backend/app/services/agri_rag.py`	Knowledge RAG	Scientific ICAR retrieval engine matching farmer queries to vetted Package of Practices (POP) documents with relevance scoring.
`backend/app/services/agri_knowledge_base.py`	Agronomy Data	Curated scientific database of crop diseases (leaf curl, blight, fruit rot, thrips) with chemical active ingredients, dosages, and organic methods.
`backend/app/services/enam_rules.py`	Legal Protection	Codified APMC trade regulations, statutory 0% commission guidelines, and farmer dispute resolution guidance (Hotline: 1800-270-0224).
`backend/app/services/farmer_db.py`	State & Memory	Dual-layer database engine. Manages farmer profiles in Supabase PostgreSQL with local SQLite resilience and conversational onboarding.
`backend/app/services/dialogue_manager.py`	Session State	Tracks rolling multi-turn conversational dialogue state, vehicle selections, active crops, and compiles WhatsApp daily crux receipts.
`backend/app/services/voice_gateway.py`	WebSocket Session	Manages full-duplex WebSocket connections, real-time barge-in audio interruption, and audio frame dispatch to the browser.
`whatsapp_baileys/server.js`	WhatsApp Microservice	Node.js service using `@whiskeysockets/baileys` with multi-device QR authentication for silent background advisory delivery.
`frontend/app.js & style.css`	Client Experience	Pure Vanilla JS/CSS client with Web Audio API visualizer, full-duplex WebSocket management, and ChatGPT-style interactive voice dock.

> 📂 **Seamless Zero-Configuration Deployment**
> The application is completely self-contained. Executing `python backend/run.py` pre-warms live government market data, initializes the local SQLite database, loads the offline agronomic knowledge base, and serves the glassmorphic frontend on `http://localhost:8000/app/` .

6. Multi-Tier Fallback Hierarchy & Resilience Matrix

To guarantee zero downtime in remote rural villages with unstable 2G/4G connectivity, Kisan-Vaani operates on a multi-tier fallback architecture:

> **Resilient 4-Tier Fallback Hierarchy**
> **Tier 1 (High-Speed Dynamic):** Groq Qwen 3.8 / OpenAI OSS (sub-1s inference) for rapid conversational voice synthesis.
> **Tier 2 (Sovereign Primary):** Gnani Evon v3.3 (`luminenio/gnani-evon-v3.3-chat`) for native Indic cultural and trade reasoning.
> **Tier 3 (Cloud Backup):** Google Gemini 2.0 Flash for multi-turn conversational resilience.
> **Tier 4 (Edge Dynamic Synthesizer):** Built-in generative context synthesizer operating locally with zero external API dependencies.

7. Operational Verification & Proven Test Scenarios (100% Dynamic)

The system has been evaluated across authentic rural dialects, noisy background conditions, and complex agronomic disputes:

SCENARIO	SPOKEN FARMER UTTERANCE	DYNAMIC GENERATIVE OUTPUT (NO PRE-FED STRINGS)
Soil & Fertilizer	*"Meri mitti me nitrogen ki kami hai, kaun si khad daalu?"*	Recommends 50 kg/ha Urea or ammonium nitrate split doses, balanced with potash and phosphorus, advising soil testing for precise field calibration.
Mandi Arbitrage	*"Bhaiya, mere paas 40 katta pyaaz hai bechne ko Bolero gaadi se."*	Computes live APMC rates, factors zero freight for own Bolero, calculates gross ₹2,68,864 payout, and explains why the top-rate mandi yields the highest net profit.
Crop Health Diagnosis	*"Mirchi ke patte upar ki taraf mud rahe hain aur peele pad rahe hain."*	Diagnoses Leaf Curl Virus and Thrips vector. Recommends organic Neem oil (5ml in 2L water) or Imidacloprid (0.05%) with a 10-day pre-harvest waiting interval.
Illegal Commission Defense	*"Aadhatiya bol raha hai 6 percent commission katega."*	Cites APMC statutory law: farmer commission is 0%. Explains that buyers must pay brokerage, warning against unauthorized deductions and providing helpline 1800-270-0224.
Domain Guardrail	*"IPL me aaj kaun sa match hai?"*	Refuses gracefully in spoken Hindi: explains expertise is dedicated to farming and mandis, and politely invites the farmer to ask a crop-related question.